

[10T] :- QB for UT1

- ✓ 1] What is IOT? Explain IOT blocks in detail. List IOT challenges.
- ✓ 2] Explain types of sensors and actuators with examples and neat diagram.
- ✓ 3] Smart Things : Layer 1 - Sensors and Actuators Layer.
- ✓ 4] Data vs Network Analytics for IOT network.
- 5] Categories of IOT application protocol and transport methods.
- 6] IOT applications in Cities & Comparison of Azure IOT and Google Cloud IOT.
- ✓ 7] Seven layer IOT reference model (IOTWF).
- ✓ 8] Enabling IOT Technologies.
- ✓ 9] Explain Raspberry Pi with diagram.
- 10] Gateways and backhaul sub-layers.
- 11] Adapting SCADA for IP.
- 12] What is SOC? Explain applications.
- 13] Health & lifestyle domain-specific IOT.

Skip QNO :- 10, 11, 12, 13.

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* Answer for QB :-

Q7 What's IOT? Explain IOT blocks in details. List IOT challenges.

→ • IOT :

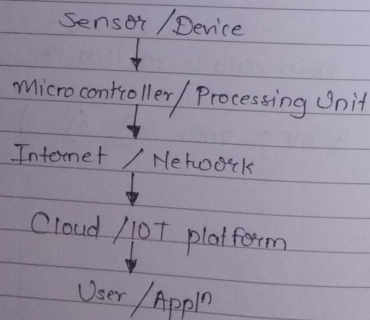
IOT is a technology in which physical object such as sensors, machines, vehicles, home appliance and devices are connected to the Internet. So that they can collect, share and exchange data automatically without human involvement.

The Smart devices sense their surrounding and send data to perform actions.

Eg., SMART AC turns on when room temperature increases.

SMART WATCH sends heart-rate data to your phone.

• IOT Block diagram :



1. Sensor / Device :

These collect data from the Environment.
eg., Temperature sensors, Camera
Gas sensor, motion sensor

2. Microcontroller / Processing Unit :

These is the brain of the IOT device.
eg., Arduino function • Raspberry PI • ESP8266

- Receive data from sensor
- Process the data
- Send it to the Internet.
- Control actuators.

3. Connectivity :

These sends data from the device to the Cloud technologies
eg., WiFi, GSM, Bluetooth, 4G/5G.

- without connectivity IOT device cannot communicate.

4. Cloud / IOT platform :

This store and process data.

Function : - Data Storage - Decision making
- Data analysis

eg., Google cloud, AWS, IOT, Microsoft, Azure

5. User Interface / Application :

This is what human use to interact with IOT system.
eg., - mobile apps, web dashboard, smart home app.

- User can
- view sensor data
 - Control device
 - Receive alerts

• IOT challenges :

- 1) Security - Hackers can steal or control devices weak password and unencrypted data are dangerous.
- 2) Privacy - IOT devices collect personal data like a location, healths, and habits.
- 3) Data Management - Huge amount of data are generated, which is difficult to store & process.
- 4) Connectivity Issue - Poor internet can cause device failure.
- 5) Cost - IOT devices, sensors and maintainance are expensive.

Q.7 Explain types of sensors and Actuators with eg and neat diagram.

→ IJ Types of Sensors :-

A sensor is a device that detect physical changes in the environment & converts them into Electrical signals that a microcontroller can read.

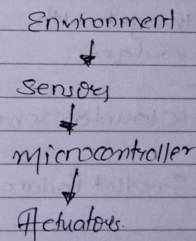
Sensor type	Function	Example
i) Temperature	Measure heat	LM35
ii) Humidity	Measure moisture	DHT11
iii) Motion	Detects movement	PIR
iv) Gas	Detects gas leakage	MQ-2
v) Light	Measure light	LDR
Sound	Measure noise level	Smart alarms

IIJ Types of Actuators :-

- An actuator is a device that performs action based on commands from the controller. It convert Electrical signal into Physical action

Actuator type	Function	Example
i) Motor	Rotates object	Water pump
ii) Servomotor	precure moment	Robotics arm
iii) solenoid	linear output	Door lock
iv) Relay	Switches appliances	Turn on/off AC
v) LED	Light output	Smart lighting
vi) Buzzer	Sound alert	Alarm system
vii) Heater	Heat	Smart heater

• Diagram :-



Eg. In a smart home

- Temperature sensors detect heat
- Microcontroller checks the value
- If high - sends signal.

Q7 Data vs Network analysis for IOT network.

parameter	Data Analysis	Network analysis
Meaning	Analyse a Data generated by IOT devices	Analyse a data flows in IOT network.
focus	What the data mean	How the data is transmitted.
Main goal	to extract useful information & pattern	to improve network performance and reliability.
Type of data used	sensors reading, Image health, data, etc.	Network traffic, bandwidth, delay, packet loss
Work on	Cloud or server	Links & devices
Uses	Predict failure	Reduce congestion
Eg.,	predicting weather using sensor data.	finding slower tailed IOT devices.

Q8 Smart Things: Layer 1 - Sensors and Actuators layer.
In IOT, smart things are physical object that can sense, process and act. They form the lowest and also most important layer.

This is also called as perception layer.

It is the physical interface b/w the real world and IOT system.

It perform two main function

- 1) Sensing
- 2) Acting

1) sensors - It collect data from physical environment and convert it into Electrical signal.

Eg.,	Sensor	Measure
	Temperature sensor	Heat level
	Humidity sensor	Moisture level
	Light sensor	Light intensity
	Motion sensor	Movement
	Gas sensor	Harmful gases
	Pressure sensor	Pressure

Action :

- Detect real world conditions
- Send data to Microcontroller.

1) **Actuators** - It performs action based on commands received from the system.
They convert Electrical Signal into Physical Signal Action.

Fig. 1	Actuators	Action
	Motor	Rotate
	Fan	Air movement
	Relay	Switch ON/OFF
	LED	Light

• Working of Layer 1

- 1) sensor detect changes in Environment.
- 2) Data is sent to the Controller.
- 3) Based on decision, actuator perform action.

Fig. 2 Temperature sensor detect high heat
Controller process data.
fan (actuator) turns ON.

• Importance of layer 1:

- i) collects real-time data.
- ii) Enable all motion.
- iii) Make device smart.

Q7 Explain Raspberry Pi with diagram.

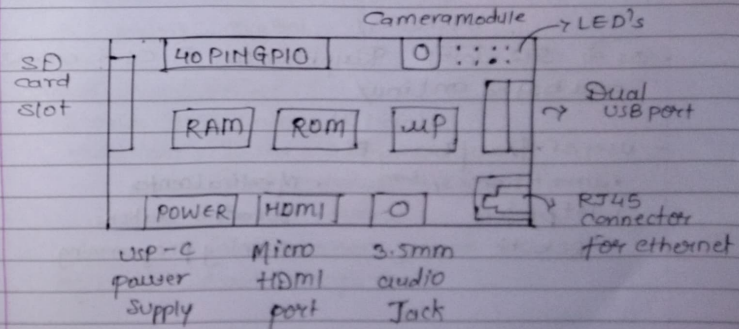
- Raspberry Pi is a single-board Computer used for IoT prototyping and development.
- Raspberry Pi is a smallest, low cost, single board computer that can perform almost all the functions of a normal computer.
- It runs on operating system (usually Linux) and can be connected to keyboard, monitor, display, sensors and internet.

- Micro controller = Single task (Eg., Camera, speaker)
- Micro processor = Multiple task.

- Raspberry Pi is based on Micro processor.

- Features of Raspberry Pi :

- Small size
- Low cost
- Support Linux
- Low power consumption.



- Main components of Raspberry Pi :-

- 1) Dual USB Port - for USB connection
- 2) Ethernet port - RJ45 connector for ethernet
- 3) SD card slot - Acts as secondary memory and stores OS and data.
- 4) RAM - Primary memory and stores running data
- 5) ROM - for startup memory
- 6) uP - Microprocessor which executes programs.
- 7) Camera Module - aka camera slot.
- 8) HDMI - Micro HDMI port which supports 4k display & connected monitor.
- 9) Power - USB C-type power supply
- 10) 40 PIN GPIO - Can connect sensors and actuators to the board
- 11) Audio - 3.5mm audio Jack where you can connect headphones, mike.

- OS के इंफ़ॉर्मेशन Raspbian OS; 2) OS is open source and based on Linux

- Uses of Raspberry Pi

- smart home system
- 1st project
- Robotic
- Medical center
- Weather stations
- Learning programming

Q7

Enabling IoT Technologies

→ They allow devices to sense, communicate, process and act intelligently.

1) sensors and Actuators :

These interact with the physical world,

- sensors collect data

- actuators perform actions

eg., Temperature sensor + fan actuator in smart AC.

2) Embedded Systems :

Small computer inside IoT devices.

eg., They control sensors, process data and send it to the cloud

eg., Arduino, Raspberry Pi, ESP8266

3) Communication Technologies :

Allow devices to connect to the internet.

Without connectivity, IoT cannot work

eg., WiFi, Bluetooth, GSM,

4G/5G, ZigBee, LoRaWAN

4) Cloud computing :

Stores and process IoT data.

functions : Data storage

Data analysis

Remote access

Device management

eg., AWS, Azure, Google Cloud.

5) Big data & Analytics:

IoT produces huge amount of data Big data tools.

process large data, find pattern, make predictions

eg., Predictions machine failure in factor.

6) Artificial Intelligence:

making IoT system smart

AI helps in face recognition, voice commands, predictive analysis, automation

eg., Smart Cameras that recognize people.

7) Cyber Security:

Protect IoT devices and data includes Encryption, Authentication, firewalls prevent hacking & drafts theft.

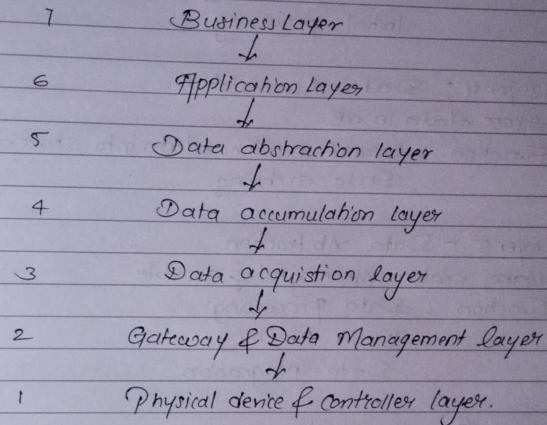
8) Web & Mobile Application:

Allow users to view and control devices

eg., Smart home apps, health monitoring apps.

Q7] Seven Layer IoT reference Model.

→ The Seven Layer IoT reference Model explains how data flows from Physical device to Business system in an IoT Environment



- Layer 1 - Physical Device & Controller
 - This layer contains sensors and actuators
 - Function: collect data, Perform action

- Layer 2 - Gateway & Data Management
 - Act as a bridge between devices and network
 - Functions: Data filtering, protocol conversion, security, Data routing

- Layer 3 - Data Acquisition
 - collect raw data from device
 - Function : Data aggregation
Data formatting
Initial processing
- Layer 4 : Data Accumulation
 - Store data in dB.
 - Function : convert realtime data into store data.
Data archiving.
- Layer 5 : Data Abstraction
 - Make data meaningful & usable
 - Function : Data Processing
Data Meaning
Data Integration
- Layer 6 : Application Layer
 - function : display info.
Sends control & commands
- Layer 7 - Business Layer
 - Manage business rule & decision
 - function : Analysis
Reports
Business logic
Automation